Prodigy Anesthesia Quick Review

Geriatric Anatomy, Physiology, and Pathophysiology

Technically, what is the difference between the terms 'elderly', 'geriatric', and 'aged'?	In scientific literature, 'elderly' and 'geriatric' are synonymous terms that both describe patients over the age of 65. The term 'aged' refers to patients over the age of 80. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 277.*
How does renal mass change with age?	The mass of the renal cortex decreases by 20-25% with age. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 893.*
What is primarily responsible for the decrease in brain mass with age?	The loss in mass is primarily due to a loss of gray matter neurons in the cortex, particularly the frontal lobes. White matter is not significantly altered. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1220.*
How is the blood circulatory time affected by aging? How does this affect intravenous and inhalation induction times?	The blood circulatory time is prolonged in the elderly. As a result, intravenous induction will progress more slowly as it takes longer for the administered drugs to reach their target sites. Inhalation induction, however, will proceed more quickly as the reduced blood flow leads to less uptake of the anesthetic and the FA/FI ratio will approach 1 more quickly. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1215.*
How is blood volume affected by aging?	Blood volume decreases with age. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1221.*
How does the decline in overall endocrine function in the elderly affect anesthetic management?	Typically, it does not. The exception may be that elderly patients tend to have higher insulin requirements perioperatively, even when a diagnosis of diabetes is not present. They exhibit a lower insulin response to glucose loads as well as increased insulin resistance in skeletal muscle. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 894.*
How is renal sodium management affected in the elderly? Why?	In elderly patients, the kidneys are not as effective at eliminating excess sodium, nor are they as effective at retaining sodium. This is due primarily to a reduction in the secretion of aldosterone. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 894.*

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Do normal, age-related changes in hepatic function require dose-adjustments for medications in the elderly?	Yes. Age-related decreases in hepatic function can result in a prolongation of the effects of typically used anesthetic agents. In particular, the plasma clearance of opiates is reduced. Careful titration, and in some cases, smaller doses may be warranted to prevent delayed awakening. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1219.*
What surgical procedure has the highest risk for postoperative delirium in elderly patients?	Although postoperative delirium can occur following any type of surgical procedure, it is most common in orthopedic surgery and cardiac surgery. Up to 60% of elderly patients undergoing orthopedic surgery experience delirium and up to 32% of patients undergoing cardiac surgery have reported symptoms of delirium. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1220.*
How does glomerular filtration rate change with age?	The glomerular filtration rate decreases about 1 mL/min each year after age 40. This can reduce renal excretion of drugs to a level where dosing adjustments need to be made by age 60. By age 80, over half of the glomeruli are lost. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 893-894.*
How is CNS activity affected by age?	CNS activity decreases with age. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1220.*
Why is thermoregulation impaired in elderly patients?	They have a lower basal metabolic rate, a higher incidence of hypothyroidism, a high proportion of body surface area to body mass, and their ability to produce peripheral vasoconstriction to reduce heat loss is not as effective. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1215.*
How does the thermoregulation differ between younger and older patients clinically? What are the advantages and disadvantages of these differences?	Temperature regulation is impaired in elderly patients. The advantage is that shivering can increase oxygen consumption by 400% and lead to hypoxia, acidosis, and cardiopulmonary complications rapidly in elderly patients. The disadvantages are that drug metabolism can be impaired with hypothermia, as can platelet function, and the risk for infection is greater. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1215.*
How do plasma protein levels change with age?	Serum albumin decreases slightly with age, but alpha-1 acid glycoprotein levels are usually unchanged or even experience a slight increase. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 893.*

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How does total body water change with age? Where does most of this change occur?	Total body water decreases with age, primarily due to the decrease in cellular water from the loss of muscle mass and increase in adipose tissue. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 893.*
At what age does functional organ performance peak?	Typically, functional organ reserve peaks at age 30 and then begins a gradual decline from that point on. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 893.*
How do liver mass and hepatic blood flow change with age?	Liver mass decreases with age and accounts for most of the 20-40% decrease in hepatic blood flow with age. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 893.*
How does the response to beta-adrenergic agents change in the elderly?	Because they have a decreased response to beta-adrenergic stimulation, the response to drugs that rely on beta-adrenergic stimulation is diminished. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 911.*
Why are geriatric patients more susceptible to hypotension under anesthesia?	Geriatric patients are at a higher risk for intraoperative hypotension due to dehydration which can result from a diminished thirst mechanism, chronic use of diuretics, and a decreased capacity for sodium and water conservation by the kidneys. Limited responsiveness to beta-adrenergic drugs and decreased cardiac contractility that is exacerbated by anesthesia drugs also contribute to the likelihood of hypotension under anesthesia. There is also a decreased ability of the heart rate to increase in response to hypovolemia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 651.*
What is the incidence of postoperative cognitive dysfunction in elderly patients?	Postoperative cognitive dysfunction occurs in 10-15% of all elderly patients undergoing anesthesia. It is not related to perioperative hypotension or hypoxia and results in a varying inability to perform simple mental tasks. It typically resolves spontaneously within three months. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 651.*

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What factors increase the risk of osteoporosis in the elderly?	Calcium and vitamin D deficiency, excessive alcohol intake, cancer, immobilization, male hypogonadism, estrogen deficiency, and chronic corticosteroid administration are all known risk factors that increase the likelihood of osteoporosis. Postmenopausal hormone replacement therapy, calcium and vitamin D supplementation, weight-bearing exercise, and calcitonin administration are all known effective ways to treat or prevent osteoporosis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 645.*
How does the risk for autoimmune disorders change with age?	The risk of autoimmune disorders increases with age. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 645.*
How do laryngeal reflexes change with advanced age?	Protective laryngeal reflexes such as coughing and swallowing decrease with increasing age, resulting in pulmonary inflammation and loss of alveolar surface area due to repeated microaspirations and contamination of the lower airway with enteric organisms. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 644.*
Elderly patients have an increased peripheral vascular resistance. Why is this and what are the eventual effects?	The increase in peripheral vascular resistance in the elderly occurs primarily due to progressive loss of arterial distensibility and leads to concentric left ventricular hypertrophy. The cardiac output may decrease due to myocardial fiber atrophy, but as the increase in afterload is greater than the decrease in cardiac output, hypertension usually results. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1215.*
How is brain mass affected by aging?	Brain mass decreases (as much as 30% by age 80). *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1220.*
How are plasma norepinephrine levels different in the elderly?	Plasma norepinephrine levels are increased in elderly patients. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1220.*
How do cerebral blood flow and CMRO2 change with age?	As CNS activity and neuronal density decrease, so does cerebral blood flow and CRMO2. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1220.*
How does the heart rate response to hypoxia and hypovolemia change with aging?	Similar to infants, the elderly have a diminished tachycardic response to hypotension, hypovolemia, or hypoxia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1215.*

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What are the major age-related changes in the cardiovascular system?	There is a decreased response to beta-receptor stimulation, the myocardium stiffens, the veins and arteries lose elasticity, sympathetic activity increases, parasympathetic activity decreases, and the conduction system may become fibrotic leading to conduction abnormalities. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 897.*
How do closing volume and closing capacity compare to functional residual capacity in the anesthetized elderly patient? What is the result?	The closing volume is increased in the elderly patient. Functional residual capacity also increases with age due to an increase in residual volume. Under anesthesia, the closing capacity is greater than the functional residual capacity, making the patient prone to atelectasis in dependent areas of the lungs. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1217.*
How is total lung capacity affected by age? Forced vital capacity? FEV1? Closing volume?	The total lung capacity decreases slightly or is unchanged. The loss of elastic recoil around the alveoli results in a decrease in the forced vital capacity and FEV1. The closing volume increases. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1217.*
How would you expect the alveolar surface in the lungs to change with age? What accounts for this change?	By age 70, the alveolar surface area available for gas exchange reduces by about 15%. The cause is an overall reduction in elastic tissue in the lungs. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1217.*
How is MAC affected by age?	MAC decreases by about 4% per decade after the age of 40. *Hines RL, Marschall KE. Stoeltings Anesthesia & Co-Existing Diseases. 6th ed. New York, NY: Churchill-Livingston; 2012: 650.*
How do renal function tests change with age?	Due primarily to fibrotic changes and decreased mass in the renal cortex, glomerular filtration rate and creatinine clearance decrease with age. The BUN increases gradually with age. Even though creatinine clearance decreases, the reduced muscle mass and creatinine production associated with increased age result in an unchanged serum creatinine level. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 911.*
How does pulse pressure change with age?	The pulse pressure is widened in the geriatric population due to an increase in the systolic pressure compared to the diastolic pressure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1215.*

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How are catecholamine levels affected by age?	Geriatric patients exhibit increased levels of circulating catecholamines *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1215.*

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